PMA Note

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Contents

1 The Real and Complex Number Systems

1.1 Introduction

Example 1.1. It is well known that there is no rational p such that $p^2 = 2$.

Define $A \coloneqq \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$ and $B \coloneqq \{p \in \mathbb{Q}^+ \mid p^2 > 2\}$. Then A contains no largest number and B contains no smallest.

Proof. It suffices to show that

$$\forall p \in A,\ \exists q \in A\ s.t.\ p < q$$

and

$$\forall p \in B,\ \exists q \in B\ s.t.\ q < p.$$

For every $p \in \mathbb{Q}^+$, take

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \tag{1}$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \tag{2}$$

Hence,

$$\begin{aligned} p \in A &\implies p^2 - 2 < 0 \\ &\implies p < q \text{ and } q^2 < 2 && (\because (1), (2)) \\ &\implies q \in A \end{aligned}$$

and

$$\begin{aligned} p \in B &\implies p^2 - 2 > 0 \\ &\implies 0 < q < p \text{ and } q^2 > 2 && (\because (1), (2)) \\ &\implies q \in B. \end{aligned}$$

We have shown the desired result. □

1.2 Ordered Sets

Definition 1.2. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

(i) If $x \in S$ and $y \in S$ then one and only one of the statements

$$x < y, \qquad x = y, \qquad y < x$$

is true.

(ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

Definition 1.3. An *ordered set* is a set S in which an order is defined.

Definition 1.4. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E.

Lower bounds are defined in the same way (with $\geq$ in place of $\leq$).

Definition 1.5. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

(i) α is an upper bound of E.

(ii) If $\gamma < \alpha$ then γ is not an upper bound of E.

Then α is called the *least upper bound of* E [that there is at most one such α is clear from (ii)] or the *supremum of* E, and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E.

Definition 1.6. An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S.

Remark 1.7. The ordered set $\mathbb{Q}$ does not have the least-upper-bound property.

The following theorem shows that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

Theorem 1.8. *Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B. Then*

$$\alpha = \sup L$$

exists in S, and $\alpha = \inf B$.

In particular, $\inf B$ exists in S.

Proof. Since B is bounded below, L is nonempty. Let $x \in B$. Then $y \leq x$ for every $y \in L$. Thus, every $x \in B$ is an upper bound of L. Since L is a nonempty subset of S that is bounded above, by hypothesis,

$$\alpha = \sup L$$

exists in S.

If $\gamma < \alpha$ then γ is not an upper bound of L, hence $\gamma \notin B$. It follows that $\alpha \leq x$ for every $x \in B$. Thus, $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L.

We conclude that α is a lower bound of B (since $\alpha \in L$) and if $\alpha < \beta$ then β is not a lower bound of B (since $\beta \notin L$), which implies $\alpha = \inf B$. □

Remark 1.9. Fix $x \in B$. Then $y \leq x$ for all $y \in L$. Thus, $\sup L \leq x$. Since the choice of $x \in B$ was arbitrary, we have

$$\sup L = \alpha \leq x \text{ for every } x \in B,$$

which implies α is a lower bound of B. Hence, $\alpha \in L$.

1.3 Fields

Definition 1.10. A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called "field axioms" (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F.
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- (A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.
- (A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x \in F$ and $y \in F$, then their product xy is in F.
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.
- (M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.
- (M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.

Definition 1.11. An *ordered field* is a *field* F which is also an *ordered set*, such that

(i) $x + y < x + z$ if $x, y, z \in F$ and $y < z$.

(ii) $xy > 0$ if $x \in F$, $y \in F$, $x > 0$, and $y > 0$.

1.4 The Real Field

Theorem 1.12. *There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.*

Theorem 1.13.

(a) If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that

$$nx > y.$$

(b) If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.

Proof.

(a) Let $x, y \in \mathbb{R}$ and $x > 0$. Suppose to the contrary that $nx \le y$ for each $n \in \mathbb{N}$ and define

$$A \coloneqq \{nx \mid n \in \mathbb{N}\}.$$

Then A is nonempty and bounded above by y. Thus, the supremum of A exists in $\mathbb{R}$. Put $\alpha = \sup A$.

Since $x > 0$, we have $\alpha - x < \alpha$, hence, $\alpha - x$ is not an upper bound of A. Therefore, $\alpha - x < mx$ for some $m \in \mathbb{N}$. Then $\alpha < (m+1)x \in A$, which is a contradiction, since α is an upper bound of A.

(b) Since Rudin's proof of part (b) is already explicit, we omit it here. □

Theorem 1.14. *For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$.*

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof. We omit the proof here. □

Corollary. *If a and b are positive and real numbers and n is a positive integer, then*

$$(ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}}.$$

Proof. We omit the proof here. □

Lemma 1.15. [1]

(a) If $E \subset \mathbb{Z}$ has the infimum, then $\inf E \in E$.

(b) If $x \in \mathbb{R}$ then there exists an $n \in \mathbb{Z}$ such that

$$n \le x < n + 1.$$

Proof.

(a) Let $\alpha = \inf E$. Suppose to the contrary that $\alpha \notin E$. Since $\alpha + 1$ is not a lower bound of E, choose an $x_0 \in E$ such that

$$\alpha < x_0 < \alpha + 1. \tag{3}$$

Since x_0 is not a lower bound of E, choose an $x_1 \in E$ again such that

$$\alpha < x_1 < x_0. \tag{4}$$

Subtracting x_0 from (4), we obtain $\alpha - x_0 < x_1 - x_0 < 0$. Since $-1 < \alpha - x_0$ from (3), it follows that $-1 < x_1 - x_0 < 0$. This contradicts the fact that $x_1 - x_0 \in \mathbb{Z}$. We conclude that $\alpha \in E$.

[1] This lemma has been added for convenience and does not appear in the original text.

(b) Let $x \in \mathbb{R}$ and define $E := \{n \in \mathbb{Z} \mid n > x\}$. By the Archimedean Property, E is nonempty. Since E is bounded below by x, the infimum of E exists in $\mathbb{R}$. Since $E \subset \mathbb{Z}$, $\inf E$ is the least element of E by (a). Thus, we have

$$\inf E - 1 \leq x < \inf E.$$

Putting $n = \inf E - 1 \in \mathbb{Z}$, we obtain the desired inequality. $\square$

Remark 1.16 (Decimals). Let $x > 0$ be real. Let n_0 be the largest integer such that $n_0 \leq x$, by lemma 1.15. Having chosen $n_0, n_1, \ldots, n_{k-1}$, let n_k be the largest integer such that

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x.$$

Let E be the set of these numbers

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \qquad (k = 0, 1, 2, \ldots). \tag{5}$$

Notice that x is an upper bound of E and if $k \in \mathbb{N}$ then

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k + 1}{10^k}.$$

Hence, we have

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < \frac{1}{10^k} \text{ for } k \in \mathbb{N}. \tag{6}$$

Suppose $y < x$. Then by the Archimedean Property, there exists a positive integer k such that $1/10^k < x - y$. From (6), we obtain

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < x - y,$$

hence

$$y < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k}.$$

Therefore, $x = \sup E$. The decimal expansion of x is

$$n_0 \cdot n_1 n_2 n_3 \cdots . \tag{7}$$

Conversely, for any infinite decimal (7), the set E of numbers (5) is bounded above, and (7) is the decimal expansion of $\sup E$.

1.5 Exercises

Exercise 1. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A).$$

Proof. Since A is bounded below, the infimum of A exists in $\mathbb{R}$. If $a \in A$ then $\inf A \leq a$. Hence,

$$-a \leq -\inf A \text{ for all } a \in A, \tag{8}$$

and it follows that $-A$ is bounded above. Thus, $\sup(-A)$ exists in $\mathbb{R}$. Since $-\inf A$ is an upper bound of $-A$ from (8), we have $\sup(-A) \leq -\inf A$, hence

$$\inf A \leq -\sup(-A). \tag{9}$$

Since $-a \leq \sup(-A)$ for every $a \in A$, we obtain $-\sup(-A) \leq a$ for all $a \in A$. Therefore, $-\sup(-A)$ is a lower bound of A and

$$-\sup(-A) \leq \inf A. \tag{10}$$

Combining (9) and (10), we conclude that $\inf A = -\sup(-A)$. $\square$

Exercise 2. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If x is real, define $B(x)$ to be the set of all numbers b^t, where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x.

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y.

Proof.

(a) Let $b^{mq} = b^{np} = x$. Since n and q are positive integers and $b > 0$, by Theorem 1.14, we have

$$b^m = x^{1/q} \text{ and } b^p = x^{1/n}.$$

It follows that

$$(b^m)^{1/n} = (x^{1/q})^{1/n} = (x^{1/n})^{1/q} = (b^p)^{1/q},$$

since $[(x^{1/q})^{1/n}]^{nq} = [(x^{1/n})^{1/q}]^{nq} = x$.

(b) Let m, n, p, q be integers and $n > 0$, $q > 0$ and let $r = m/n$ and $s = p/q$. Then by the Corollary to Theorem 1.14, we obtain

$$\begin{aligned}
b^{r+s} &= b^{\frac{m}{n}+\frac{p}{q}} = b^{\frac{mq+np}{nq}} \\
&= \left(b^{mq+np}\right)^{\frac{1}{nq}} \\
&= \left(b^{mq}b^{np}\right)^{\frac{1}{nq}} \\
&= \left(b^{mq}\right)^{\frac{1}{nq}} \left(b^{np}\right)^{\frac{1}{nq}} \\
&= b^{\frac{mq}{nq}} b^{\frac{np}{nq}} = b^{\frac{m}{n}} b^{\frac{p}{q}} = b^r b^s.
\end{aligned}$$

(c) Note that by Theorem 1.14, $b^r > 0$ if $b > 1$ and r is rational.

Let n be a positive integer and let $0 < a < 1$ be real. We first show that $0 < a^{1/n} < 1$. Suppose to the contrary that $a^{1/n} \geq 1$. Then we obtain

$$1 \leq \left(a^{1/n}\right)^n = a < 1,$$

a contradiction. Therefore, if $0 < a < 1$ then $0 < a^{1/n} < 1$, where n is a positive integer.

Suppose r and s are rational. Let m, n, p, and q be integers and let n and q be positive. Then we write $r = m/n$, $s = p/q$.

Suppose $r < s$. Then

$$r - s = \frac{m}{n} - \frac{p}{q} = \frac{mq - np}{nq} < 0,$$

hence

$$0 < b^{mq-np} = \frac{1}{b^{np-mq}} < 1$$

and

$$0 < b^{r-s} = b^{\frac{mq-np}{nq}} = \left(b^{mq-np}\right)^{1/nq} < 1.$$

Thus, if $r < s$ then

$$b^r = b^{(r-s)+s} = b^{r-s}b^s < b^s.$$

If t is rational and $t \leq r$ then $b^t \leq b^r$. It follows that $B(r)$ is bounded above by b^r and $\sup B(r)$ exists in $\mathbb{R}$. Since b^r is an upper bound of $B(r)$,

$$\sup B(r) \leq b^r.$$

Since $b^r \in B(r)$,

$$b^r \leq \sup B(r).$$

We conclude that $b^r = \sup B(r)$.

(d) Let t_1 and t_2 be rational. If $t_1 \leq x$ and $t_2 \leq y$ then $t_1 + t_2 \leq x + y$. Thus, we have

$$b^{t_1}b^{t_2} = b^{t_1+t_2} \leq \sup B(x+y) = b^{x+y}.$$

Fix $t_1 \leq x$. Then

$$b^{t_2} \leq b^{x+y}/b^{t_1} \text{ for all } t_2 \leq y.$$

Hence,

$$b^y = \sup B(y) \leq b^{x+y}/b^{t_1}.$$

Since the choice of t_1 was arbitrary,

$$b^{t_1} \leq b^{x+y}/b^y \text{ for all } t_1 \leq x,$$

and it follows that

$$b^x = \sup B(x) \leq b^{x+y}/b^y.$$

Therefore, we obtain

$$b^x b^y \leq b^{x+y}. \tag{11}$$

Notice that

$$0 < b^{t_1} \leq b^x \text{ for every } t_1 \leq x$$

and

$$0 < b^{t_2} \leq b^y \text{ for every } t_2 \leq y.$$

Hence, for every $t_1 \leq x$ and $t_2 \leq y$,

$$b^{t_1+t_2} = b^{t_1}b^{t_2} \leq b^x b^y.$$

Now, to obtain

$$b^{x+y} = \sup B(x+y) \leq b^x b^y \tag{12}$$

from above inequality, we need the following statement:

$$b^x = \sup\left\{b^t \mid t \in \mathbb{Q}, t < x\right\}.$$

Let us prove it. Notice that

$$b^n - 1 = (b-1)(b^{n-1} + \cdots + 1) \geq n(b-1)$$

for any positive integer n. Put $c = b^{1/n} > 1$. Then we have

$$b - 1 \geq n(b^{1/n} - 1).$$

Hence, if $t > 1$ and $n > (b-1)/(t-1)$ then

$$\frac{n(b^{1/n}-1)}{t-1} \leq \frac{b-1}{t-1} < n,$$

and it follows that

$$b^{1/n} < t.$$

Note that if $t > 1$ then $b^{1/n} < t$ for sufficiently large n.

Let $B_0(x) = \{b^t \mid t \in \mathbb{Q}, t < x\}$. We need to show that b^x is the supremum of $B_0(x)$. It is clear that b^x is an upper bound of $B_0(x)$. Thus, it remains to prove that if $y < b^x$ then y is not an upper bound.

Suppose $0 < y < b^x$. Then $b^x/y > 1$ and it follows that

$$b^{1/n} < b^x/y$$

for sufficiently large n. Hence, we have

$$y < \frac{b^x}{b^{1/n}} < b^x.$$

By the density of rationals, choose a rational t such that $x - (1/n) < t < x$. Since $x < s < t + (1/n)$ for some rational s,

$$b^x \leq b^s < b^{t+(1/n)} = b^t b^{1/n}$$

and

$$y < \frac{b^x}{b^{1/n}} < b^t.$$

Thus, y is not an upper bound of $B_0(x)$ and therefore $b^x = \sup B_0(x)$.

Now, let t be a rational and let $t < x + y$. Pick $t_2 \in \mathbb{Q}$ such that $t - x < t_2 < y$ and set $t_1 = t - t_2$. Then for every $t \in \mathbb{Q}$ with $t < x + y$, there are rational $t_1 < x$ and $t_2 < y$ such that $t = t_1 + t_2 < x + y$, hence $b^t = b^{t_1+t_2} = b^{t_1}b^{t_2} \leq b^x b^y$. Thus, we obtain (12).

Combining (11) and (12), we conclude that

$$b^{x+y} = b^x b^y.$$

□

Exercise 3. Prove that no order can be defined in the complex field that turns it into an ordered field.

Proof. Suppose to the contrary that there exists an order $<$ on $\mathbb{C}$ which turns $\mathbb{C}$ into an ordered field. Let $\mathbf{0}$ and $\mathbf{1}$ denote the additive and multiplicative identities of the complex field, respectively. Then $\mathbf{0} = (0,0)$ and $\mathbf{1} = (1,0)$.

Note that $i = (0,1) \neq \mathbf{0}$. If $x \neq \mathbf{0}$ then $x^2 > \mathbf{0}$. It follows that

$$i^2 = (0,1)(0,1) = (-1,0) > \mathbf{0}.$$

Since $(-1,0) + (1,0) = \mathbf{0}$, $(-1,0)$ is the additive inverse of $(1,0)$. Since $\mathbf{1} > \mathbf{0}$ and if $x > \mathbf{0}$ then $-x < \mathbf{0}$ in any ordered field, we obtain

$$(-1,0) = -(1,0) = -\mathbf{1} < \mathbf{0}.$$

This contradicts the Trichotomy Property. □

2 Basic Topology

2.1 Finite, Countable, and Uncountable Sets

We omit this section here.

2.2 Metric Spaces

Definition 2.1. A set X, whose elements we shall call *points*, is said to be a *metric space* if with any two points p and q of X there is associated a real number $d(p, q)$, called the *distance* from p to q, such that

(a) $d(p, q) > 0$ if $p \neq q$; $d(p, p) = 0$;

(b) $d(p, q) = d(q, p)$;

(c) $d(p, q) \leq d(p, r) + d(r, q)$, for any $r \in X$.

Any function with these three properties is called a *distance function*, or a *metric*.

Example 2.2. $\mathbb{R}^k$ is a metric space.

Definition 2.3. By the *segment* (a, b) we mean the set of all real numbers x such that $a < x < b$.

By the *interval* $[a, b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

Occasilnally we shall also encounter "half-open intervals" $[a, b)$ and $(a, b]$; the first consists of all x such that $a \leq x < b$, the second of all x such that $a < x \leq b$.

If $a_i < b_i$ for $i = 1, \ldots k$, the set of all points $\mathbf{x} = (x_1, \ldots, x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ $(1 \leq i \leq k)$ is called a *k-cell.* Thus a 1-cell is an interval, a 2-cell is a rectangle, etc.

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open* (or *closed*) *ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| < r$ (or $|\mathbf{y} - \mathbf{x}| \leq r$).

We call a set $E \subset \mathbb{R}^K$ *convex* if

$$\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

Example 2.4. Balls and k-cells are convex.

Proof. If $|\mathbf{y} - \mathbf{x}| < r$, $|\mathbf{z} - \mathbf{x}| < r$, and $0 < \lambda < 1$, then

$$\begin{aligned} |\lambda\mathbf{y} + (1 - \lambda)\mathbf{z} - \mathbf{x}| &= |\lambda(\mathbf{y} - \mathbf{x}) + \lambda\mathbf{x} + (1 - \lambda)\mathbf{z} - \mathbf{x}| \\ &= |\lambda(\mathbf{y} - \mathbf{x}) + (1 - \lambda)(\mathbf{z} - \mathbf{x})| \\ &\leq \lambda|\mathbf{y} - \mathbf{x}| + (1 - \lambda)|\mathbf{z} - \mathbf{x}| \\ &< \lambda r + (1 - \lambda)r = r. \end{aligned}$$

Hence, balls are convex.

Let $A \subset \mathbb{R}^k$ be a k-cell and let $\mathbf{x}, \mathbf{y} \in A$ and $0 < \lambda < 1$. We write $\mathbf{x} = (x_1, \ldots x_k)$, $\mathbf{y} = (y_1, \ldots, y_k)$ with

$$a_i < b_i \text{ and } \mathbf{x}, \mathbf{y} \in [a_i, b_i] \text{ for } i = 1, \ldots, k.$$

If $x, y \in [a, b]$ with $a < b$ and $0 < \lambda < 1$ then

$$\lambda a \leq \lambda x \leq \lambda b,$$

and

$$(1 - \lambda)a \leq (1 - \lambda)y \leq (1 - \lambda)b,$$

hence

$$a \leq \lambda x + (1-\lambda)y \leq b.$$

It follows that each $[a_i, b_i]$ $(1 \leq i \leq k)$ is convex.

Therefore, every k-cell is convex. $\square$

Definition 2.5. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X.

(a) *A neighborhood* of p is a set $N_r(p)$ consisting of all q such that $d(p,q) < r$, for some $r > 0$. The number r is called the *radius* of $N_r(p)$.

(b) A point p is a *limit point* of the set E if *every* neighborhood of p contains a point $q \neq p$ such that $q \in E$.

(c) If $p \in E$ and p is not a limit point of E, then p is called an *isolated point* of E.

(d) E is *closed* if every limit point of E is a point of E.

(e) A point p is an *interior* point of E if there is a neighborhood N of p such that $N \subset E$.

(f) E is *open* if evefy point of E is an interior point of E.

(g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.

(h) E is *perfect* if E is closed and if every point of E is a limit point of E.

(i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p,q) < M$ for all $p \in E$.

(j) E is *dense in* X if every point of X is a limit point of E, or a point of E (or both).

Theorem 2.6. *Every neighborhood is an open set.*

Proof. Consider a neighborhood $E = N_r(p)$, and let q be any point of E. Then choose an $h > 0$ such that

$$d(p,q) = r - h.$$

If $s \in N_h(q)$ then

$$d(p,s) \leq d(p,q) + d(q,s) < r - h + h = r,$$

and it follows that $N_h(q) \subset E$. Thus there is a neighborhood of q that is a subset of E. Therefore, every $q \in E$ is an interior point of E. We conclude that E is an open set. $\square$

Theorem 2.7. *If p is a limit point of a set E, then every neighborhood of p contains infinitely many points of E.*

Proof. Suppose there is a neighborhood N of p which contains only a finite number of points of E. Let $q_1, \ldots, q_n$ be those points of $N \cap E$, which are distinct from p, and put

$$r = \min_{1 \leq m \leq n} d(p, q_m) > 0$$

Then $d(p, q_m) \geq r$ for $m = 1, \ldots, n$ and hence all the points q_m are not in the neighborhood $N_r(p)$. Since

$$N_r(p) \subset N$$

and hence

$$N_r(p) \cap E \setminus \{p\} \subset N \cap E \setminus \{p\} = \{q_1, \ldots, q_m\} \subset E,$$

$N_r(p)$ contains no element $q \neq p$ in E (notice $N_r(p) \cap E \setminus \{p\}$ is empty). Thus, p is not a limit point of E. This is a contradiction. $\square$

Corollary. *A finite point set has no limit points.*

Theorem 2.8. *Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α. Then*

$$\left(\bigcup_\alpha E_\alpha\right)^c = \bigcap_\alpha (E_\alpha^c)\,. \tag{13}$$

Proof. We have

$$\begin{aligned}
x \in \left(\bigcup_\alpha E_\alpha\right)^c &\implies x \notin \left(\bigcup_\alpha E_\alpha\right) \\
&\implies \forall\alpha, x \notin E_\alpha && (\because \text{ negation of the definition}) \\
&\implies \forall\alpha, x \in E_\alpha^c \\
&\implies x \in \bigcap_\alpha E_\alpha^c.
\end{aligned}$$

Hence $(\bigcup_\alpha E_\alpha)^c \subset \bigcap_\alpha (E_\alpha^c)$. Since

$$\begin{aligned}
x \in \bigcap_\alpha (E_\alpha^c) &\implies \forall\alpha, x \in E_\alpha^c \\
&\implies \forall\alpha, x \notin E_\alpha \\
&\implies x \notin \bigcup_\alpha E_\alpha && (\because \text{ negation of the definition}) \\
&\implies x \in \left(\bigcup_\alpha E_\alpha\right)^c,
\end{aligned}$$

we obtain the reverse. This establishes the theorem. □

Theorem 2.9. *A set E is open if and only if its complement is closed.*

Proof. First, suppose E^c is closed. Choose $x \in E$. Then $x \notin E^c$, and x is not a limit point of E^c, since every limit point of E^c is a point of E^c (contrapositive). Hence there exists a neighborhood N of x such that $E^c \cap N$ is empty, that is, $N \subset E$. Thus E is open.

Next, suppose E is open. Let x be a limit point of E^c. Then every neighborhood of x contains a point of E^c, hence there is no neighborhood N of x such that $N \subset E$. Thus, x is not an interior point of e and therefore $x \notin E$, since E is open. It follows that if x is a limit point of E^c then $x \notin E$, hence $x \in E^c$. Therefore, E^c is closed. □

Corollary. *A set F is closed if and only if its complement is open.*

Theorem 2.10.

(a) For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.

(b) For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.

(c) For any finite collection $G_1, \ldots G_n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.

(d) For any finite collection $F_1, \ldots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.

Proof. Put $G = \bigcup_\alpha G_\alpha$. Then

$$\begin{aligned}
x \in G &\implies \exists\alpha\, s.t.\, x \in G_\alpha \\
&\implies x \text{ is an interior point of } G_\alpha \\
&\implies \exists r_0 > 0\, s.t.\, N_{r_0}(x) \subset G_\alpha \\
&\implies \exists r_0 > 0\, s.t.\, N_{r_0}(x) \subset G.
\end{aligned}$$

Hence, every $x \in G$ is an interior point of G and it follows that G is open.

(b) follows from (a), Theorem 2.8, and Theorem 2.9. We see that

$$
\begin{aligned}
F_\alpha \text{ is closed } &\implies F_\alpha^c \text{ is open.} && (\because \text{ Theorem } 2.9)\\
&\implies \bigcup_\alpha F_\alpha^c \text{ is open.} && (\because \text{ (a)})\\
&\implies \left(\bigcap_\alpha F_\alpha\right)^c \text{ is open.} && (\because \text{ Theorem } 2.8)\\
&\implies \bigcap_\alpha F_\alpha \text{ is closed.} && (\because \text{ Theorem } 2.9)
\end{aligned}
$$

Since Rudin's proof of part (c) and (d) are already explicit, we omit it here. □